

initial-etl

Minh Ngo

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.5.1      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
loans_raw <- read_csv("../data-raw/2024-pudb.csv")
```

```
## Rows: 46981 Columns: 56
## -- Column specification -----
## Delimiter: ","
## chr   (7): Bank, FIPSSStateNumericCode, FIPSCountyCode, CensusTractIdentifier...
## dbl   (48): Year, LoanCharacteristicsID, CoreBasedStatisticalAreaCode, Census...
## date  (1): PrepaymentPenaltyExpirationDate
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
# Check missingness for borrower 2
borrower2_check <- loans_raw %>%
  mutate(
    b2_credit_missing = (Borrower2CreditScoreValue == 9),
    b2_age_missing = Borrower2AgeAtApplicationYears %in% c(998, 999),

    # define "borrower 2 exists"
    b2_exists = !(b2_credit_missing & b2_age_missing)
  )

# Summary
borrower2_check %>%
  summarise(
    total_rows = n(),
```

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    borrower2_exists = sum(b2_exists, na.rm = TRUE),
    pct_exists = mean(b2_exists, na.rm = TRUE),

    usable_credit = sum(!b2_credit_missing, na.rm = TRUE),
    pct_credit_available = mean(!b2_credit_missing, na.rm = TRUE),

    usable_age = sum(!b2_age_missing, na.rm = TRUE),
    pct_age_available = mean(!b2_age_missing, na.rm = TRUE)
  )

```

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## # A tibble: 1 x 7
##   total_rows borrower2_exists pct_exists usable_credit pct_credit_available
##   <int>         <int>         <dbl>         <int>         <dbl>
## 1     46981         26948         0.574         26781         0.570
## # i 2 more variables: usable_age <int>, pct_age_available <dbl>

```

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cols_to_keep <- c(
  # IDs / context
  "Year",
  "LoanCharacteristicsID",
  "Bank",
  "FIPSSStateNumericCode",
  "FIPSCountyCode",
  "CoreBasedStatisticalAreaCode",

  # Neighborhood / area context
  "CensusTractMinorityRatioPercent",
  "CensusTractMedFamIncomeAmount",
  "LocalAreaMedianIncomeAmount",
  "HUDMedianIncomeAmount",

  # Core borrower / loan financials
  "TotalMonthlyIncomeAmount",
  "LTVRatioPercent",
  "NoteRatePercent",
  "NoteAmount",
  "HousingExpenseRatioPercent",
  "TotalDebtExpenseRatioPercent",
  "Borrower1CreditScoreValue",
  "Borrower2CreditScoreValue",
  "PMICoveragePercent",

  # Loan characteristics
  "LoanPurposeType",
  "ProductCategoryName",
  "MortgageType",
  "ScheduledTotalPaymentCount",
  "LoanAmortizationMaxTermMonths",
  "MortgageLoanSellerInstType",
  "PropertyUsageType",
  "PropertyUnitCount",
  "PropertyType",

```

```

# Borrower characteristics
"BorrowerCount",
"BorrowerFirstTimeHomebuyer",
"Borrower1AgeAtApplicationYears",
"Borrower2AgeAtApplicationYears",
"Borrower1SexType",
"Borrower2SexType",
"Borrower1Race1Type",
"Borrower2Race1Type",
"EmploymentBorrowerSelfEmployed"
)

fhfa_master <- loans_raw %>%
  select(any_of(cols_to_keep))

head(fhfa_master)

## # A tibble: 6 x 37
##   Year LoanCharacteristicsID Bank FIPSSStateNumericCode FIPSCountyCode
##   <dbl> <dbl> <chr> <chr> <chr>
## 1 2024 2820897 Boston 33 017
## 2 2024 2820898 Boston 23 031
## 3 2024 2820899 Boston 23 007
## 4 2024 2820900 Boston 33 011
## 5 2024 2820901 Boston 25 017
## 6 2024 2820902 Boston 23 005
## # i 32 more variables: CoreBasedStatisticalAreaCode <dbl>,
## # CensusTractMinorityRatioPercent <dbl>, CensusTractMedFamIncomeAmount <dbl>,
## # LocalAreaMedianIncomeAmount <dbl>, HUDMedianIncomeAmount <dbl>,
## # TotalMonthlyIncomeAmount <dbl>, LTVRatioPercent <dbl>,
## # NoteRatePercent <dbl>, NoteAmount <dbl>, HousingExpenseRatioPercent <dbl>,
## # TotalDebtExpenseRatioPercent <dbl>, Borrower1CreditScoreValue <dbl>,
## # Borrower2CreditScoreValue <dbl>, PMICoveragePercent <dbl>, ...

write_csv(fhfa_master, "../data-clean/pudb_reduced.csv")

fhfa_clean <- fhfa_master %>%
  mutate(
    # Credit scores
    Borrower1CreditScoreValue = na_if(Borrower1CreditScoreValue, 9),
    Borrower2CreditScoreValue = na_if(Borrower2CreditScoreValue, 9),

    # Age
    Borrower1AgeAtApplicationYears = na_if(Borrower1AgeAtApplicationYears, 999),
    Borrower1AgeAtApplicationYears = na_if(Borrower1AgeAtApplicationYears, 998),
    Borrower2AgeAtApplicationYears = na_if(Borrower2AgeAtApplicationYears, 999),
    Borrower2AgeAtApplicationYears = na_if(Borrower2AgeAtApplicationYears, 998),

    # Race
    Borrower1Race1Type = ifelse(Borrower1Race1Type %in% c(6,7), NA, Borrower1Race1Type),
    Borrower2Race1Type = ifelse(Borrower2Race1Type %in% c(6,7), NA, Borrower2Race1Type),

    # Sex

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Borrower1SexType = ifelse(Borrower1SexType %in% c(3,4,6), NA, Borrower1SexType),
Borrower2SexType = ifelse(Borrower2SexType %in% c(3,4,6), NA, Borrower2SexType)
)

```

```

fhfa_clean <- fhfa_clean %>%
  mutate(
    TotalMonthlyIncomeAmount = as.numeric(TotalMonthlyIncomeAmount),
    LTVRatioPercent = as.numeric(LTVRatioPercent),
    NoteRatePercent = as.numeric(NoteRatePercent),
    TotalDebtExpenseRatioPercent = as.numeric(TotalDebtExpenseRatioPercent),
    NoteAmount = as.numeric(NoteAmount)
  )

```

```
summary(fhfa_clean)
```

```

##      Year      LoanCharacteristicsID      Bank      FIPSSStateNumericCode
## Min.   :2024    Min.   :2813828      Length:46981      Length:46981
## 1st Qu.:2024    1st Qu.:2825573      Class :character    Class :character
## Median :2024    Median :2837318      Mode  :character    Mode  :character
## Mean   :2024    Mean   :2837318
## 3rd Qu.:2024    3rd Qu.:2849063
## Max.   :2024    Max.   :2860808
##
## FIPSCountyCode      CoreBasedStatisticalAreaCode
## Length:46981      Min.   :10100
## Class :character    1st Qu.:19820
## Mode  :character    Median :31140
##                      Mean   :38998
##                      3rd Qu.:41860
##                      Max.   :99999
##
## CensusTractMinorityRatioPercent CensusTractMedFamIncomeAmount
## Min.   : 0.000      Min.   : 0
## 1st Qu.: 8.707      1st Qu.: 79595
## Median :14.345      Median : 97072
## Mean   :19.910      Mean   :103678
## 3rd Qu.:25.218      3rd Qu.:120375
## Max.   :99.889      Max.   :250001
##
## LocalAreaMedianIncomeAmount HUDMedianIncomeAmount TotalMonthlyIncomeAmount
## Min.   : 0      Min.   : 26500      Min.   : 0
## 1st Qu.: 82833      1st Qu.: 87300      1st Qu.: 7016
## Median : 91784      Median : 97300      Median : 10708
## Mean   : 92303      Mean   : 98142      Mean   : 13494
## 3rd Qu.: 99766      3rd Qu.:105400      3rd Qu.: 15881
## Max.   :172303      Max.   :180400      Max.   :999999
##
## LTVRatioPercent NoteRatePercent NoteAmount      HousingExpenseRatioPercent
## Min.   : 1.00      Min.   :3.125      Min.   : 18000      Min.   : 0.00
## 1st Qu.: 69.00      1st Qu.:6.250      1st Qu.: 188000      1st Qu.:16.90
## Median : 80.00      Median :6.625      Median : 283000      Median :22.79
## Mean   : 75.23      Mean   :6.534      Mean   : 311321      Mean   :23.28
## 3rd Qu.: 89.00      3rd Qu.:6.875      3rd Qu.: 400000      3rd Qu.:29.31

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## Max. :110.00 Max. :8.375 Max. :1474400 Max. :71.77
##
## TotalDebtExpenseRatioPercent Borrower1CreditScoreValue
## Min. : 0.00 Min. :1.000
## 1st Qu.:27.21 1st Qu.:4.000
## Median :34.70 Median :5.000
## Mean :33.89 Mean :4.472
## 3rd Qu.:41.04 3rd Qu.:5.000
## Max. :85.13 Max. :5.000
## NA's :159
## Borrower2CreditScoreValue PMICoveragePercent LoanPurposeType
## Min. :1.000 Min. : 0.000 Min. :1.000
## 1st Qu.:4.000 1st Qu.: 0.000 1st Qu.:1.000
## Median :5.000 Median : 0.000 Median :1.000
## Mean :4.526 Mean : 7.182 Mean :1.824
## 3rd Qu.:5.000 3rd Qu.:12.000 3rd Qu.:2.000
## Max. :5.000 Max. :51.000 Max. :6.000
## NA's :20200
## ProductCategoryName MortgageType ScheduledTotalPaymentCount
## Length:46981 Min. :0.00000 Min. : 85
## Class :character 1st Qu.:0.00000 1st Qu.:360
## Mode :character Median :0.00000 Median :360
## Mean :0.06203 Mean :348
## 3rd Qu.:0.00000 3rd Qu.:360
## Max. :3.00000 Max. :360
##
## LoanAmortizationMaxTermMonths MortgageLoanSellerInstType PropertyUsageType
## Min. : 85 Length:46981 Min. :1.00
## 1st Qu.:360 Class :character 1st Qu.:1.00
## Median :360 Mode :character Median :1.00
## Mean :348 Mean :1.09
## 3rd Qu.:360 3rd Qu.:1.00
## Max. :360 Max. :2.00
##
## PropertyUnitCount PropertyType BorrowerCount
## Min. :1.000 Length:46981 Min. :1.000
## 1st Qu.:1.000 Class :character 1st Qu.:1.000
## Median :1.000 Mode :character Median :2.000
## Mean :1.009 Mean :1.588
## 3rd Qu.:1.000 3rd Qu.:2.000
## Max. :4.000 Max. :5.000
##
## BorrowerFirstTimeHomebuyer Borrower1AgeAtApplicationYears
## Min. :0.0000 Min. :18.00
## 1st Qu.:0.0000 1st Qu.:31.00
## Median :0.0000 Median :40.00
## Mean :0.2588 Mean :42.82
## 3rd Qu.:1.0000 3rd Qu.:53.00
## Max. :1.0000 Max. :97.00
##
## Borrower2AgeAtApplicationYears Borrower1SexType Borrower2SexType
## Min. :18.00 Min. :1.000 Min. :1.000
## 1st Qu.:31.00 1st Qu.:1.000 1st Qu.:1.000
## Median :40.00 Median :1.000 Median :2.000

```

##	Mean	:42.86	Mean	:1.345	Mean	:1.688
##	3rd Qu.	:53.00	3rd Qu.	:2.000	3rd Qu.	:2.000
##	Max.	:98.00	Max.	:2.000	Max.	:2.000
##	NA's	:20033	NA's	:2005	NA's	:21318
##	Borrower1Race1Type	Borrower2Race1Type	EmploymentBorrowerSelfEmployed			
##	Min.	:1.000	Min.	:1.000	Min.	:0.0000
##	1st Qu.	:5.000	1st Qu.	:5.000	1st Qu.	:0.0000
##	Median	:5.000	Median	:5.000	Median	:0.0000
##	Mean	:4.803	Mean	:6.264	Mean	:0.1217
##	3rd Qu.	:5.000	3rd Qu.	:8.000	3rd Qu.	:0.0000
##	Max.	:5.000	Max.	:8.000	Max.	:1.0000
##	NA's	:4141	NA's	:2543		